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Subject: Programming with Java

ASSIGNMENT No : 11

TITLE : Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Outcome

Exercise

1. Create an interface **Printable** and implement it in Student and Employee classes.

```
3  J AS6EX1.java 2   J ShapeDemo.java 3   J Ecommerce.ja   J AS9E
J PrintableDemo.java > PrintableDemo
interface Printable {
    void print
class Student
extends Object
implements Printable
}

class Student implements Printable {

    public void print() {
        System.out.println("Student Details Displayed");
    }
}

class Employee implements Printable {

    public void print() {
        System.out.println("Employee Details Displayed");
    }
}

public class PrintableDemo {
    Run main | Debug main
    public static void main(String[] args) {

        Student s = new Student();
        Employee e = new Employee();

        s.print();
        e.print();
    }
}
```

OUTPUT-

```
PROBLEMS 14   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
Student Details Displayed
Employee Details Displayed
```

2. Create an interface **Switchable** with a method turnOn(). Implement the interface in **Light** and **Fan** classes to display the device status.

```java
interface Switchable {
    void turnOn();
}

class Light implements Switchable {

    public void turnOn() {
        System.out.println("Light is ON");
    }
}

class Fan implements Switchable {

    public void turnOn() {
        System.out.println("Fan is ON");
    }
}

public class SwitchableDemo {
    public static void main(String[] args) {

        Light l = new Light();
        Fan f = new Fan();

        l.turnOn();
        f.turnOn();
    }
}
```

OUTPUT-

```
Light is ON
Fan is ON
```

Conclusion

The program was successfully implemented to demonstrate the **implementation of interfaces** in Java. It showed how interfaces provide abstraction by defining a common set of methods that implementing classes must implement. This practical enhanced the understanding of abstraction, multiple inheritance of type, and code reusability, while promoting modular, flexible, and maintainable object-oriented applications.